Finding Parameters That Make Piecewise Functions Continuous

Solving for the constant that makes a piecewise function continuous, filling a removable hole with the limit value, and showing when no value can work

Directions.

- f is continuous at $x = p$ exactly when all three of these hold: $f(p)$ exists, $\lim_{x\to p} f(x)$ exists, and the two are equal.
- At a junction between two pieces, "the limit exists" means the left-hand and right-hand limits agree. Set them equal and solve for the unknown constant.
- Show the one-sided limits explicitly before you solve. An answer with no limit statement earns no credit on the AP exam.
- Give exact values (fractions, not decimals).

1. Let

$$f(x) = \begin{cases} 3x + a, & x < 2 \\ x^2 + 1, & x \geq 2 \end{cases}$$

Find the value of a that makes f continuous at $x = 2$.

Answer: ________________

2. Let

$$f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

Find the value of k that makes f continuous at $x = 3$.

Answer: ________________

3. An oven's temperature, in degrees Celsius, t minutes after it is switched on is modelled by

$$T(t) = \begin{cases} 4t + c, & 0 \leq t < 5 \\ 32 - 2t, & t \geq 5 \end{cases}$$

Find the value of c, in degrees Celsius, that makes T continuous at $t = 5$.

Answer: ________________

4. Let

$$f(x) = \begin{cases} x^2, & x \leq 1 \\ ax + b, & 1 < x < 4 \\ 10, & x \geq 4 \end{cases}$$

Find the values of a and b that make f continuous on all of $(-\infty, \infty)$.

Answer: ____________________

5. Let

$$f(x) = \begin{cases} \dfrac{\sqrt{x+7} - 3}{x - 2}, & x \neq 2 \\ k, & x = 2 \end{cases}$$

Find the exact value of k that makes f continuous at $x = 2$. (Rationalize the numerator first.)

Answer: ____________________

6. Let

$$f(x) = \begin{cases} \dfrac{e^{4x} - 1}{2x}, & x \neq 0 \\ k, & x = 0 \end{cases}$$

Find the value of k that makes f continuous at $x = 0$.

Answer: ________________

7. Let

$$f(x) = \begin{cases} 2x + a, & x < 3 \\ x^2 + b, & 3 \leq x < 5 \\ 4x + 1, & x \geq 5 \end{cases}$$

Find the values of a and b that make f continuous everywhere. Start with the junction that involves only one unknown.

Answer: ________________

8. A student claims that setting $k = 5$ makes

$$f(x) = \begin{cases} \dfrac{1}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

continuous at $x = 3$. Using one-sided limits, explain why *no* value of k makes f continuous there, and name the type of discontinuity at $x = 3$.

Answer: ____________________